

ass9_mnish7331

March 17, 2026

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[97]: import seaborn as sns
import matplotlib.pyplot as plt
```

```
[99]: df = sns.load_dataset('titanic')
```

```
[101]: print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class \
0	0	3	male	22.0	1	0	7.2500	S	Third
1	1	1	female	38.0	1	0	71.2833	C	First
2	1	3	female	26.0	0	0	7.9250	S	Third
3	1	1	female	35.0	1	0	53.1000	S	First
4	0	3	male	35.0	0	0	8.0500	S	Third

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

```
[103]: print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   class       891 non-null    category
9   who         891 non-null    object
```

```
10  adult_male    891 non-null    bool
11  deck          203 non-null    category
12  embark_town  889 non-null    object
13  alive         891 non-null    object
14  alone         891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
```

```
[105]: df['age'] = df['age'].fillna(df['age'].median())
```

```
[107]: df['survived'] = df['survived'].map({0: 'Did Not Survive', 1: 'Survived'})
```

```
[109]: sns.set_theme(style="whitegrid")
```

```
[111]: plt.figure(figsize=(10, 6))
sns.boxplot(
    x='sex',
    y='age',
    hue='survived',
    data=df
)

plt.title('Age Distribution by Gender and Survival')
plt.xlabel('Gender')
plt.ylabel('Age')

plt.show()
```

